

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied business analytics

Practical – I

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Step 1: Import Libraries

Python

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Load Dataset

Python

```
df = pd.read_csv("ecomer.csv")
```

Step 3: Exploratory Data Analysis

Initial Data Inspection

Python

```
# Display first 5 records
df.head()
```

	order_id	order_date	customer_id	product_category	region	quantity	unit_price
0	10001	1/1/2022	1102	Beauty	South	7	373.65
1	10002	1/2/2022	1435	Clothing	South	7	47.74
2	10003	1/3/2022	1860	Beauty	East	3	311.28
3	10004	1/4/2022	1270	Electronics	West	5	524.47
4	10005	1/5/2022	1106	Clothing	West	5	139.87

Python

```
# Display last 5 records
df.tail()
```

	order_id	order_date	customer_id	product_category	region	quantity
4995	14996	9/5/2035	1289	Clothing	West	2
4996	14997	9/6/2035	1294	Beauty	North	3
4997	14998	9/7/2035	1450	Beauty	West	3
4998	14999	9/8/2035	1903	Electronics	South	4
4999	15000	9/9/2035	1214	Home	East	7

Dataset Structure and Metadata

Python

```
# Shape, Columns, and Info
print(df.shape)
print(df.columns)
df.info()
print(df.dtypes)
```

```
Index(['order_id', 'order_date', 'customer_id', 'product_category', 'region',
       'quantity', 'unit_price', 'discount', 'payment_method', 'delivery_days',
       'customer_rating', 'revenue'],
      dtype='str')
```

```
<class 'pandas.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   order_id              5000 non-null  int64
1   order_date            5000 non-null  str
2   customer_id           5000 non-null  int64
3   product_category      5000 non-null  str
4   region                5000 non-null  str
5   quantity              5000 non-null  int64
6   unit_price            5000 non-null  float64
7   discount              5000 non-null  float64
8   payment_method        5000 non-null  str
9   delivery_days         5000 non-null  int64
10  customer_rating        5000 non-null  float64
11  revenue                5000 non-null  float64
dtypes: float64(4), int64(4), str(4)
memory usage: 468.9 KB
```

```
order_id      int64
order_date    str
customer_id    int64
product_category  str
region        str
quantity      int64
unit_price    float64
discount      float64
payment_method str
delivery_days int64
customer_rating float64
revenue       float64
dtype: object
```

(5000, 12)

Data Quality Check

Python

```
# Check for missing and duplicate values
print(df.isnull().sum())
print(f"Duplicates: {df.duplicated().sum()}")
```

```
order_id      0
order_date    0
customer_id   0
product_category 0
region        0
quantity      0
unit_price    0
discount      0
payment_method 0
delivery_days 0
customer_rating 0
revenue       0
dtype: int64
```

Step 4: Statistical Summary & Categorical Analysis

Python

```
# Descriptive Statistics
df.describe()
```

	order_id	customer_id	quantity	unit_price	discount	delivery_days
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000
mean	12500.500000	1505.701200	4.044800	308.418774	0.179984	6.118800
std	1443.520003	290.836902	2.020398	169.259369	0.101404	3.153264
min	10001.000000	1000.000000	1.000000	15.150000	0.000000	1.000000
25%	11250.750000	1253.000000	2.000000	161.895000	0.090000	3.000000
50%	12500.500000	1510.000000	4.000000	309.890000	0.180000	6.000000
75%	13750.250000	1761.000000	6.000000	455.557500	0.270000	9.000000
max	15000.000000	1999.000000	7.000000	599.960000	0.350000	11.000000

Python

```
# Categorical value counts
print(df["product_category"].value_counts())
print(df["region"].value_counts())
print(df["payment_method"].value_counts())
```

order_id	5000		
order_date	5000		
customer_id	989		
product_category	4		
region	4		
quantity	7		
unit_price	4788		
discount	36		
payment_method	3		
delivery_days	11		
customer_rating	41		
revenue	4940		
dtype: int64			

product_category		
Electronics	1777	
Clothing	1531	
Home	969	
Beauty	723	
Name: count, dtype: int64		

region		
West	1289	
North	1276	
East	1227	
South	1208	
Name: count, dtype: int64		

payment_method		
Card	2270	
COD	1774	
Wallet	956	
Name: count, dtype: int64		

<StringArray>		
['Beauty', 'Clothing', 'Electronics', 'Home']		
Length: 4, dtype: str		

Python

```
# Revenue metrics
print(f"Average Revenue: {df['revenue'].mean()}")
print(f"Max Revenue: {df['revenue'].max()}")
print(f"Min Revenue: {df['revenue'].min()}")
```

np.float64(1021.955148)

np.float64(4119.33)

np.float64(11.21)

Step 5: Visualizations & Correlation

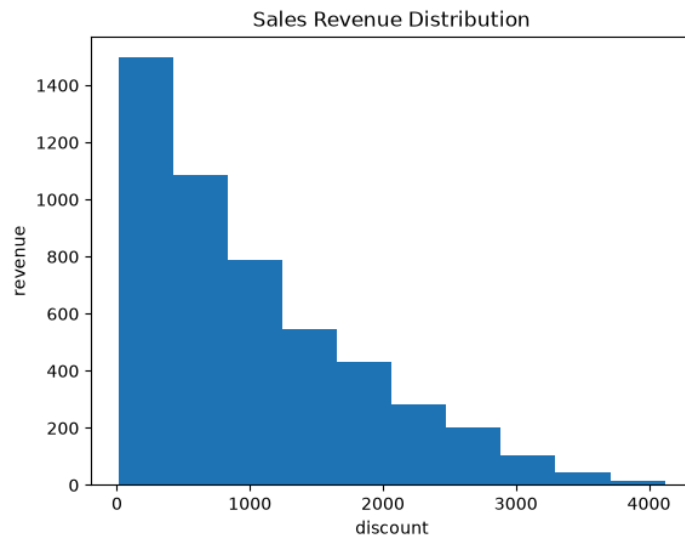
Python

```
# Correlation Analysis
df.corr(numeric_only=True)
```

	order_id	customer_id	quantity	unit_price	discount	delivery_days
order_id	1.000000	0.017504	-0.003368	0.017876	0.011110	0.006196
customer_id	0.017504	1.000000	0.023097	0.009748	-0.012617	-0.017440
quantity	-0.003368	0.023097	1.000000	0.001498	-0.004302	-0.008748
unit_price	0.017876	0.009748	0.001498	1.000000	0.013215	0.015623
discount	0.011110	-0.012617	-0.004302	0.013215	1.000000	-0.000382
delivery_days	0.006196	-0.017440	-0.008748	0.015623	-0.000382	1.000000
customer_rating	0.004984	0.020302	0.019168	0.005025	0.012776	-0.017625
revenue	0.009681	0.025499	0.623564	0.678032	-0.139296	0.005444

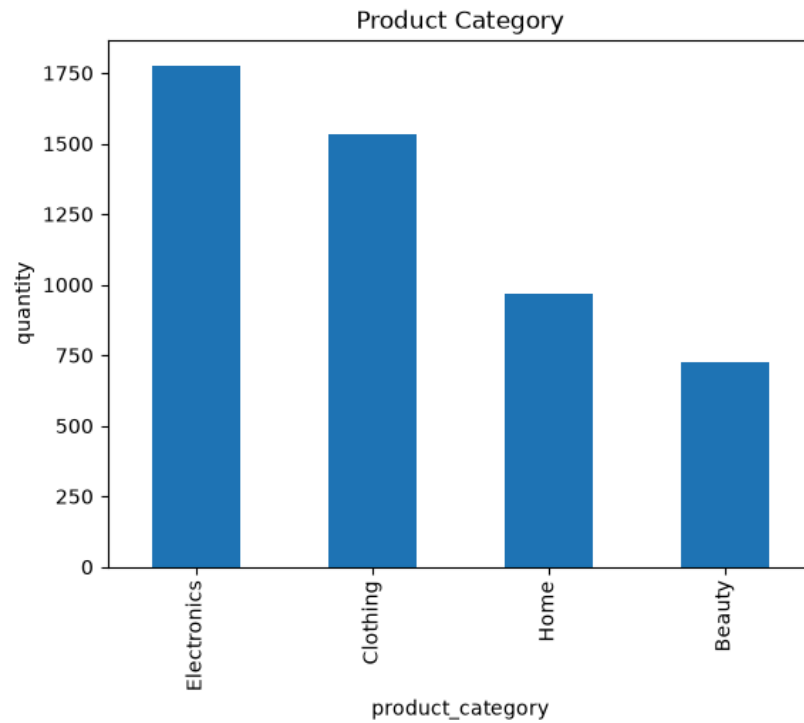
Python

```
# Revenue Distribution Histogram
plt.hist(df["revenue"])
plt.title("Sales Revenue Distribution")
plt.xlabel("discount")
plt.ylabel("revenue")
plt.show()
```



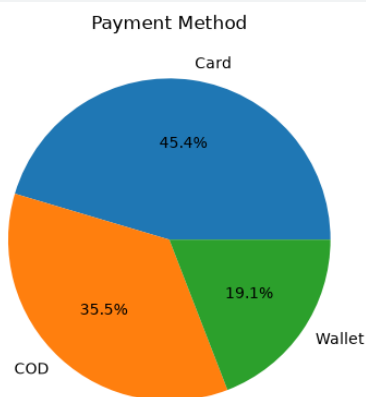
Python

```
# Product Category Bar Chart
df["product_category"].value_counts().plot(kind="bar")
plt.title("Product Category")
plt.xlabel("product_category")
plt.ylabel("quantity")
plt.show()
```



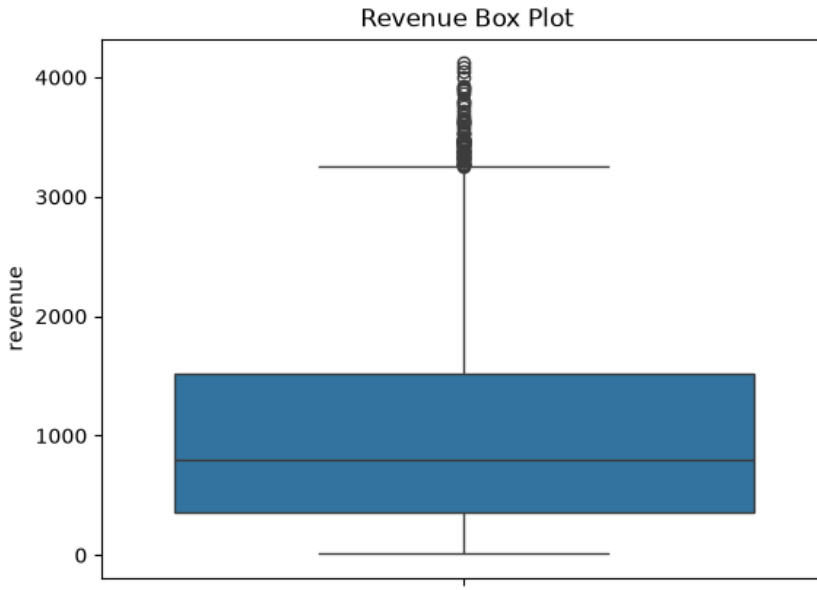
Python

```
# Payment Method Pie Chart
df["payment_method"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Payment Method")
plt.ylabel("")
plt.show()
```



Python

```
# Revenue Box Plot
sns.boxplot(y=df["revenue"])
plt.title(" Revenue Box Plot")
plt.show()
```



Python

```
# Correlation Heatmap
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Blues")
plt.title("Correlation Heatmap")
plt.show()
```

